

Appendix A Technical Description of the Agent-Based Simulation Model

A.1 Purpose of the Model

The Agent-Based Simulation (ABS) model was developed to improve understanding of how innovation capability emerges within Micro and Small Enterprises (MSEs). Consistent with the structural modeling perspective proposed by Troost et al. (2023), the objective of the model is explanatory rather than predictive. The model seeks to reduce uncertainty regarding the mechanisms through which organizational conditions influence individual capability development and how these processes generate organizational innovation capability trajectories.

The model assumes that organizational innovation capability emerges from the cumulative development of individual innovation capabilities and therefore adopts a microfoundational perspective in which workers constitute the primary agents of innovation capability development.

A.2 Agents and Environment

Workers

Workers constitute the principal agents of the simulation.

Each worker:

- Evaluates future opportunities.
- Allocates effort among alternative activities.
- Learns from experience.
- Interacts with other workers.
- Accumulates innovation capability through time.

Workers differ in:

- Potential Well-Being.
- Innovation Profile.
- Innovation capability levels.
- Interaction histories.

This heterogeneity allows the model to represent different developmental trajectories among workers operating in the same organizational environment.

Organizational Environment

The environment is represented through NetLogo patches. Rather than representing physical space, patches represent organizational states associated with different levels of innovation capability.

Each patch contains an attribute called: advantage, which represents the innovation capability level associated with a specific organizational state.

Workers interact with these organizational states through their own: advantage, which represents individual innovation capability.

This representation provides a visual mechanism for observing how workers move toward organizational states associated with higher or lower innovation capability levels through time.

A.3 Organizational Mechanisms

The model incorporates three organizational mechanisms.

Organizational Support

Organizational Support is represented as an organizational condition associated with actions intended to encourage innovation-oriented behavior and capability-development activities.

- Reinforces innovation-oriented behavior.
- Facilitates learning opportunities.
- Provides knowledge and guidance.
- Promotes developmental activities.

Consistent with intrapreneurship perspectives (Blanka, 2019), such actions may encourage workers to take initiative, experiment, and voluntarily participate in organizational improvement processes. In the present model, these actions are represented as a general organizational condition that supports innovation-oriented behavior rather than as a detailed support mechanism.

Governance

Governance represents the degree of autonomy and participation available to workers. Higher levels of governance increase opportunities for initiative, decision-making, and experimentation. Lower levels constrain worker discretion and restrict innovation-oriented behavior.

Innovation culture contagion

Innovation Culture Contagion represents the transmission of innovation-oriented attitudes, values, and behavioral dispositions among workers. Through interaction and organizational communication processes, workers influence one another's perceptions regarding innovation-oriented behavior.

A.4 Individual Variables

Potential Well-Being (PWB).

Potential Well-Being represents workers' perceptions regarding the future benefits associated with innovation-oriented behavior. Workers compare current conditions with possible future outcomes and evaluate whether changing behavior may generate better results. The variable is grounded in behavioral economics and bounded rationality assumptions.

Innovation Profile (IP)

Innovation Profile represents the worker's disposition toward:

- learning,
- experimentation,
- cooperation,
- initiative,
- self-development.

IP influences behavioral priorities but does not directly represent innovation capability.

Ladvantage

Ladvantage represents the worker's accumulated individual innovation capability.

It evolves through:

- learning,
- cooperation,
- interaction,
- experience accumulation.

Higher values indicate higher levels of individual innovation capability accumulated by the worker.

A.5 Activity Portfolio

During each simulation cycle, workers allocate effort among alternative activities that compete for limited attention and time. These activities represent the principal channels through which workers generate organizational value, develop innovation capabilities, interact with colleagues, and acquire new knowledge. Consequently, the allocation of effort among activities constitutes one of the central behavioral mechanisms of the model. Table A1 summarizes the six activities available to workers and their role within the simulation.

Table A1. Activities Available for Worker Time Allocation

Activity	Description
Productive Work	Routine operational activities required for organizational functioning

Self-Cultivation	Learning and capability-development activities
Cooperation	Knowledge sharing and collaborative work
Interaction with the External Innovation Environment	Relationships with external actors such as customers, suppliers, universities, and innovation-support institutions
Low-Value Activities	Activities generating limited learning or innovation benefits
Self-Oriented Activities	Activities directed toward personal objectives rather than organizational development

The model assumes that innovation capability does not emerge directly from organizational conditions but through the activities in which workers invest their available effort. Activities such as self-cultivation, cooperation, and interaction with the external innovation environment contribute positively to capability accumulation, whereas low-value and self-oriented activities reduce opportunities for capability development. As a result, differences in effort allocation become an important source of divergence in innovation capability trajectories.

A.6 Behavioral Logic

The behavioral logic of the model is based on the assumption that organizational innovation capability emerges from the interaction between organizational conditions, worker perceptions, behavioral dispositions, and capability-development activities. Workers do not develop innovation capability directly as a consequence of organizational conditions. Instead, organizational conditions influence perceptions regarding future opportunities, which subsequently affect innovation-oriented dispositions and effort-allocation decisions. Through repeated participation in learning, cooperation, and interaction activities, workers accumulate innovation capability over time.

In Figure A1 Organizational conditions influence Potential Well-Being (PWB), which affects Innovation Profiles (IP). Innovation Profiles guide workers' time allocation decisions, determining participation in capability-development activities. Through repeated learning and interaction processes, workers accumulate individual innovation capability (Ladvantage), from which organizational innovation capability emerges.

Figure A1. Behavioral Logic of the Agent-Based Model



Source: Authors' elaboration.

A.7 Experimental Design

The simulation was implemented in NetLogo and executed using the BehavioralSpace experimentation environment. Experimental scenarios were generated through combinations of Organizational Support, Governance, and Innovation Culture Contagion. Multiple simulation runs were performed under each configuration to identify recurring developmental patterns rather than isolated outcomes. Output data were exported for statistical analysis using R. Table A2 summarizes the principal characteristics of the simulation environment, experimentation process, and analytical procedures used in the study.

Table A2. Experimental Design and Analysis Configuration

Component	Value
Modeling Platform	NetLogo
Experimentation Tool	BehavioralSpace
Statistical Environment	R

Number of Workers	4
MSE Definition Adopted	Organizations with fewer than 20 employees
Simulation Runs	65,536
Number of Ticks	80
Experimental Configurations	32,768
Replications per Configuration	2
Total Temporal Observations Generated	5,242,880
Organizational Mechanisms	Organizational Support, Governance, Innovation Culture Contagion
Individual Variables	Potential Well-Being (PWB), Innovation Profile (IP), Ladvantage
Main Analytical Outputs	Ladvantage, Organizational Innovation Capability, Capability Trajectories
Validation Approach	Structural consistency, plausibility, and explanatory adequacy
Horizon of Analysis	80 simulation ticks

The experimental design was structured as a full-factorial exploration of the parameter space. Fifteen experimental variables were evaluated at two levels, generating 32,768 unique parameter configurations. Each configuration was executed twice, resulting in 65,536 simulation runs. Simulations were executed for 80 ticks, generating more than five million temporal observations for subsequent analysis in R. Although the study adopts the definition of MSEs as organizations employing fewer than 20 workers, a population of four workers was used during large-scale experimentation. Preliminary experiments indicated that the principal capability-development mechanisms and trajectory patterns remained qualitatively stable across different workforce sizes within the range considered for MSEs. Consequently, a reduced population size was selected to maintain computational feasibility while preserving the explanatory objectives of the study.

A.8 Statistical Analysis

Simulation outputs were analyzed through:

- descriptive statistics,
- longitudinal exploration,
- clustering techniques for capability-development trajectories.

Following recommendations derived from statistical consultancy work conducted with senior Statistics students from the Universidad Nacional de Colombia, trajectory-oriented

approaches were prioritized because innovation capability evolves through time and exhibits strong temporal dependence.

The analysis revealed three recurring trajectories:

1. Accumulation
2. Sustaining
3. Erosion

These trajectories were identified through longitudinal clustering techniques and subsequently interpreted as accumulation, sustaining, and erosion paths of innovation capability development. These trajectories constitute the principal outputs reported in the article.

A.9 Validation Strategy

Validation followed the explanatory and structural approach proposed by Troost et al. (2023). The objective was not predictive accuracy but plausibility, coherence, and explanatory adequacy.

Validation included:

- consistency with innovation capability theory,
- consistency with microfoundational perspectives,
- consistency with behavioral economics assumptions,
- internal logical coherence,
- plausibility assessment through comparison with observations obtained from innovation, productivity, entrepreneurship, and business-development activities conducted with MSEs within the Colombian National Learning Service (SENA).

Consistent with Troost et al. (2023), the model was considered valid to the extent that it generated plausible mechanisms capable of explaining how innovation capability develops and evolves within MSEs.